MIL-E-1/1551A (USAF)
27 JANUARY 1972
SUPERSEDING
MIL-E-1/1551 (USAF)
19 November 1964

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 5682

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode
See figure 1
Mounting position: Vertical anode down
Weight: 50 pounds nominal

ABSOLUTE RATINGS: F1 = 30 MHz; F2 = 88 MHz (56 percent of ratings)

Parameter:	Ef	Ebb	Ec	Ib	Ic	Pg
Unit:	V	kVdc	kVdc	Adc	Adc	W
Maximum:						
B Audio:	16.5 ±5%	16.0	--	18	---	2000
B Telep:	16.5 ±5%	16.0	--	14	---	2000
C Telep:	16.5 ±5%	10.0	-3.2	14	4.0	2000
C Teleg:	16.5 ±5%	16.0	-3.2	20	4.0	2000
TEST CONDITIONS:	16.5	----	--	--	---	--

ABSOLUTE RATINGS:

Parameter:	Pi	Pp	T (Seal)	Mod	Anode Cooling
Unit:	kW	kW	°C	---	---
Maximum:					
B Audio:	280	120	165	---	Water
B Telep:	150	120	165	---	Water
C Telep:	140	80	165	Anode	Water
C Teleg:	300	120	165	---	Water
TEST CONDITIONS:	---	---	(Note 1)	---	(Note 2)

GENERAL:

Qualification: Not Required: Substitute First Article Tests (See Note 4)

Holding Period: t = 24 hours

(A) Complete Revision

Type 5682

FSC 5960

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:57Z
Check the source to verify that this is the current version before use.

METHOD	TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						Min	Max	
	Quality conformance inspection, part 1							
1266	Gas current	Ebb = 16.0 kVdc Ec/Ib = 7.5 Adc t = 300 sec. (note 3)	0.65	II	Ic	---	-900	µAdc
1261	Electrode voltage (Grid)	Ebb = 16.0 kVdc Ec/Ib = 0.020 Adc	0.65	II	Ec	-520	-780	Vdc
1261	Electrode voltage (1) (anode)	Ec = -100 Vdc Ebb/Ib = 3.0 Adc	0.65	II	(1)Ebb	4.6	6.0	kVdc
1261	Electrode voltage (2) (anode)	Ec = 0 Vdc Ebb/Ib = 3.0 Adc	0.65	II	(2)Ebb	1.8	3.0	kVdc
1236	Power oscillation	Ebb = 22.5 kVdc Ec = -2500 Vdc Rf drive/Ib = 20Adc Ic = 3.0 Adc ±10% F = 13.6 MHz ±10% t = 15 minutes	0.65	II	Po	270	---	kW (useful)
1231	Peak emission	eb = ec = 3000 v	0.65	II	is	280	---	a
	Quality conformance inspection, part 2							
1301	Filament current		----	--	If	315	335	A
1331	Direct-interelectrode capacitance	No shield	----	--	Cgp Cgk Cpk	78 100 2.0	92 122 3.2	pF pF pF
1105	Permanence of marking		----	--	---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:57Z
Check the source to verify that this is the current version before use.

NOTES:

1. Forced-air cooling of grid and filament seals is required. At frequencies up to 15 MHz, approximately 250 cfm air flow should be directed primarily on filament seals; at higher frequencies or high ambient temperatures, additional air flow may be required on the grid seals. Air flow should be distributed to maintain uniform temperature, not greater than 165°C, around the circumference of the seals.

2. Anode is water cooled. Water flow shall be between 60 and 70 gpm; pressure drop across water jacket at 65 gpm is 45 psi.

3. After the specified time, note grid current as (1)Ic. Then increase bias to 0.020 Adc anode current per grid voltage test, and note grid current as (2)Ic. Record Gas current as Ic = (1)Ic -(2)Ic.

4. First Article Testing: The First Article sample furnished by the contractor/ manufacturer shall consist of preproduction tubes in the quantity indicated by Table 1 of MIL-E-1. In addition, the contractor shall furnish along with the test reports, written assurance that:

   a. The First Article sample is representative of the product to be furnished under the contract.

   b. Production articles will be fabricated in the facility that produced the sample tubes.

   c. Production articles will be fabricated from new materials.

   The contractor shall furnish a test plan to the Contracting Officer for approval before proceeding with First Article testing. The test plan shall list all production test and laboratory test facilities to be utilized in First Article testing and production phases of the contract. Upon receipt of test plan approval, the contractor shall proceed with First Article testing. First Article testing shall include all tests and requirements of this specification and conform to Table 1 of MIL-E-1 where applicable. Upon conclusion of testing, three copies of the test report prepared in accordance with MIL-STD-831 and certified by the Government Representative shall be forwarded along with the test samples to the Contracting Officer for approval.

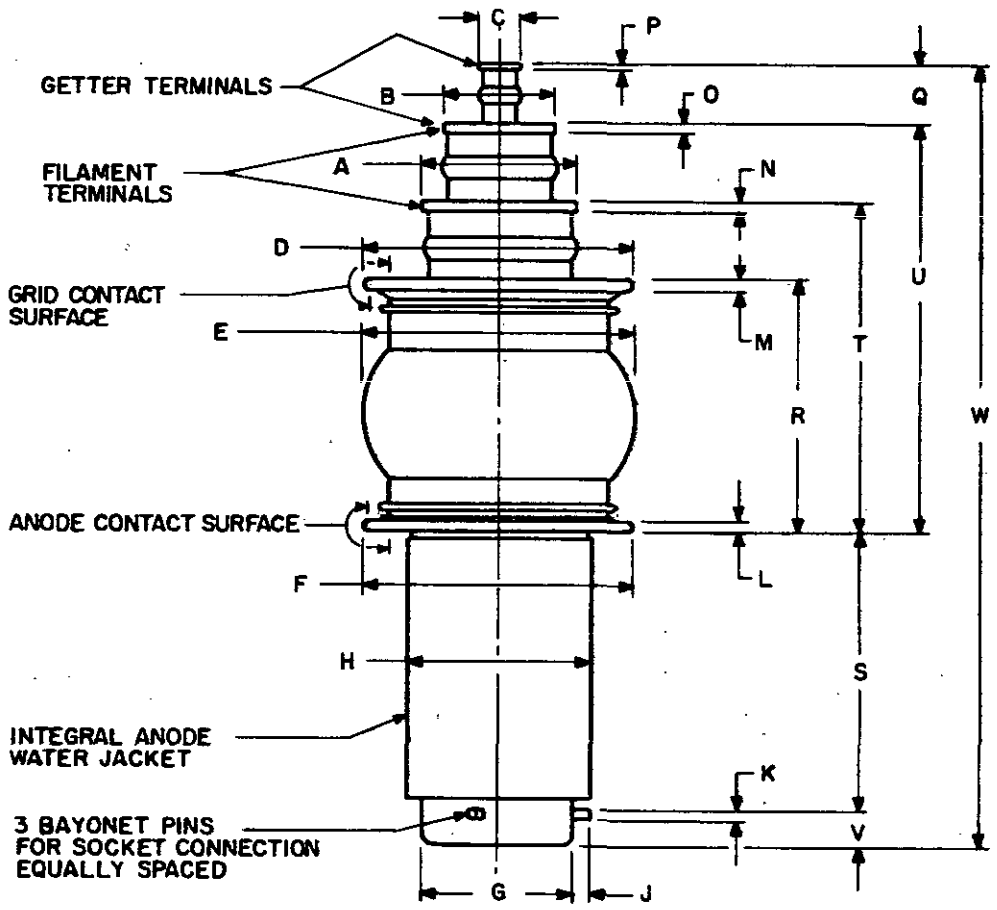
Custodian:
Air Force - 80

Preparing activity:
Air Force - 80

Review activities:
Air Force - 11, 17
DSA - ES

(Project 5960-F826-14)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:57Z
Check the source to verify that this is the current version before use.



QUALITY CONFORMANCE INSPECTION, PART 2									
DIM	INCHES		MM		DIM	INCHES		MM	
	MIN	MAX	MIN	MAX		MIN	MAX	MIN	MAX
A	4.609	4.671	117.07	118.64	M	.375	—	9.53	—
B	3.282	3.344	83.36	84.94	N	.312	—	7.92	—
C	1.219	1.281	30.96	32.54	O	.312	—	7.92	—
D	7.969	8.031	202.41	203.99	P	.187	—	4.75	
E	—	8.00	—	203.20	Q	—	1.75	—	44.45
F	7.969	8.031	202.41	203.99	R	7.343	7.593	186.51	192.86
G	4.485	4.500	113.92	114.30	S	8.000	8.250	203.20	209.55
H	—	5.50	—	139.70	T	9.656	9.906	245.26	251.61
J	.468	.531	11.89	13.49	U	11.88	12.12	301.75	307.85
K	.300	.320	7.62	8.13	V	1.03	1.09	26.16	27.69
L	.312	—	7.92	—	W	—	23.25	—	590.55

FIGURE 1 *Outline of tube type 5682*

☆ U. S. GOVERNMENT PRINTING OFFICE: 1972 -714-163/5174

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:57Z
Check the source to verify that this is the current version before use.